CARDIOLINE

Service Manual

ar2100**view**

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The information contained in this publication is subject to change without notice requirements. In case of non correspondence between user manual and Device, refer to our sales representative.

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1. Introduction

ar2100**view** is an electrocardiograph with dual power supply, (mains and rechargeable internal batteries), which in the basic configuration allows:

- ✓ recording of an ECG in either automatic, manual mode and pre-programmed;
- ✓ visualizing in real time the ECG signal on a graphic display in print format: 3 - 3+1 - 3+3 -6 - 6+6 - 12 channels;
- ✓ printing the ECG on 210 mm in different format paper using a high-resolution thermal printer;
- ✓ sorting of the tests according to clock, date and alphanumeric keyboard to manage user and patient data;
- ✓ Set up to 4 different profiles can use to adapt the operation of the various requirements;

In just a few minutes your **ar**2100**view** can be equipped with:

- ✓ Memory option: storage of up to 200 full ECG exams, with no need to print out the ECG;
- ✓ ECG measurement option: automatic ECG parameter measurement program;
- ✓ ECG analysis program: automatic ECG interpretation program;
- ✓ PC archive option: archival storage of the ECG in a personal computer running cubeecg software;
- ✓ PC ECG option: real time display of the 12 ECG leads on a personal computer screen to allow management of patient medical records and archiving of ECG's in digital format thanks to the **cube**ecg software.

For further information contact directly:

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1.1 Information and recommendations concerning the safety of use

- The electrical system used by the device must be in accordance with the standard in force.
- Always use the equipment according to the instructions in this manual.
- The device is equipped with a set of standard accessories. For reasons of safety, reliability and conformity with the Medical Devices Directive

93/42/EEC, use only original accessories or accessories approved by the manufacturer.

- The device is equipped with a special long-life thermal head writing system, which allows maximum writing precision. To avoid frequent and costly replacements and repairs, always use the original paper or paper approved by the manufacturer. The manufacturer will not accept liability for any damage to the device or any other adverse effect caused by the use of unsuitable paper.
- Do not subject the device to impact or excessive vibrations.
- Do not allow liquids to penetrate inside the device. If this should accidentally occur, have the device tested by an Authorised Assistance Centre to verify its functional efficiency, before using it again.
- Make sure that the value of the supply voltage corresponds to that indicated on the data plate of the device.
- If you are using the device in connection with others, ensure that: all connections are made by skilled persons; all connections comply with safety regulations; all other devices connected respond likewise to regulations. Non-compliance with regulations can cause physical harm to the patient connected and to the person operating the device. Should it be difficult to obtain the necessary information for assessing the risk of the individual connections, apply directly to the manufacturers concerned or avoid making the connections.
- In the event of other equipment being connected directly or indirectly to the patient, check for the possible risks caused by the sum of the leakage currents on the body of the patient.
- The device is protected against defibrillation discharges in accordance with IEC standard; to ensure that the signal is restored, use only original electrodes or electrodes responding to IEC and AAMI standards.
- If an electrosurgical scalpel is in use, the patient cable should be disconnected from the device.
- At all events, when defibrillators or high-frequency surgical devices are being used at the same time, it is essential to take the greatest care. If there is any doubt when such devices are in use, disconnect the patient from the electrocardiograph temporarily.
- The device recognises the impulses generated by a pacemaker and does not interfere with its operation, as prescribed by standards in use at the time of drafting this manual.
- Avoid exposing the equipment to extreme temperatures, excessive dust or dirt, and very salty or damp environments; observe the ambient conditions described in detail under the "*Technical specifications*" heading.
- Periodically check the efficiency of all accessories and of the device itself. Use the “test incorporated” function for a first verification of the efficiency. Contact the Authorised Assistance Centre whenever the device seems to be operating irregularly.
- To prolong the life of your device, have it checked periodically by an Authorised Assistance Centre.

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- ***Warning***: *The electrocardiograph cannot be used for intracardiac applications.*
- ***Warning***: *It is therefore necessary, before activating the equipment, to make sure of the connection to ground (normally secured by the power supply cable). If grounding of the main electrical service is not certain, do not connect the device and use it powered only by the rechargeable internal battery.*
- ***Warning***: *do not use the device in the presence of anaesthetics or volatile gases!*
- ***Warning***: *devices for medical applications must be used only by persons who by virtue of training or practical experience are able to ensure maximum safety and effectiveness in operation. Operators must in any event read this manual carefully and familiarise themselves with the instrument before using it on a patient.*
- ***Warning***: *the indications obtained using automatic interpreting programs or other diagnostic aids must be reviewed and countersigned by a qualified medical person!*
- The manufacturer will acknowledge liability for the safety, reliability and functional efficiency of the device only if:
  - modifications and repairs are performed by the manufacturer or by an Authorised Assistance Centre;
  - the AC mains power supply of the building responds to current regulations;
  - the device is operated according to user instructions;
  - any accessories in use are those approved by the manufacturer.

Guidelines for Bluetooth transmission

The device meets the requirements of the R&TTE Directive about radio equipment.
However, in order to protect the device from other instruments which don't comply with these regulations, we recommend placing the device as far as possible from other devices that use the Bluetooth transmission.

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2. Description of the device

The device consists of the following basic elements:

Mother board
This is a “Fine line” multilayer printed circuit board for mounting SMD components.
It hosts most of the electronic circuits of the device.

Battery
Battery Pack made of 10 cells NiMH 12V, 2100mA.

Display
Display assembly is made of: frame, colour LCD TFT, 480x272 pixel, effective area 95 x 54 mm, backlit with 9 led (4,3”).

Keyboard
The keyboard consists of 47 functional keys plus a Led of main supply.

Printer unit
This consists of the thermal printer head support and the mechanical elements required to position it correctly.

Bluetooth module (RS232)
The function of the Bt module is to transmit and receive data from an external PC.

Motor assembly
The DC motor with gears for paper speed. Also included strobo disk for speed feedback.

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3. Inputs and outputs

Direct connections from the **ar**2100**view** to external equipment may only be made using the BT, not by cable.

3.1 Connection to the patient cable port

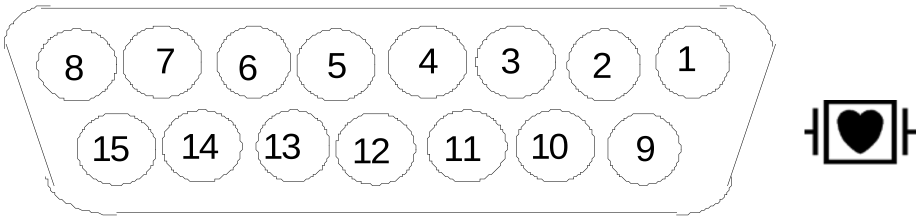

Image from connection side

Pin 1	=	IN	C2	(electrode C2)
Pin 2	=	IN	C3	(electrode C3)
Pin 3	=	IN	C4	(electrode C4)
Pin 4	=	IN	C5	(electrode C5)
Pin 5	=	IN	C6	(electrode C6)
Pin 6	=	AGND		(analog ground)
Pin 7	=	NC		(not connected)
Pin 8	=	DGND		(digital ground)
Pin 9	=	IN	R	(electrode R)
Pin 10	=	IN	L	(electrode L)
Pin 11	=	IN	F	(electrode F)
Pin 12	=	IN	C1	(electrode C1)
Pin 13	=	NC		(not connected)
Pin 14	=	IN	N	(electrode N)
Pin 15	=	NC		(not connected)

3.2 USB Port

ar2100**view** is equipped with a mini USB plug with the function to update the firmware. USB port is located in the battery compartment, refer to the manual "Cardioline Loader" for detailed instructions on updating the firmware of the device.

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4. Testing the safety characteristics

The safety regulations envisage two important tests:

✓ *The leakage currents test* measures the value of the currents lost in relation to the safety of the patient and the operator.

Warning: *All safety tests must be performed according to standards EN.60601-1, EN 60601-2-25 unless otherwise specified in the local safety regulations.*

4.1 Leakage currents test

Warning: *This test must be performed every time the device has been opened for inspection and/or repair, and in any event every two years, unless otherwise specified by the local safety regulations.*

Connect the electrocardiograph to the battery charger, and then connect this assembly to the measuring instrument according to the instrument's manual, recalling that:

1) *The leakage current to the casing* is measuring between the mains supply circuits and a metal sheet no greater than 20 x 10 cm pressed against the casing of the device.
2) *The leakage current in the patient* is measured between the mains and the applied part. For connection to the applied part use the patient lead itself.
3) *The leakage current in the patient with mains voltage directly on the applied part* (first failure condition) is measured between the meta sheet connected to the device and the applied part.
4) *The auxiliary current in the patient* is measured singly on each electrode (excluding the reference electrode) compared to all the other electrodes connected together.

Note: *Make the measurements following the indications in the instrument user manual, and check that the leakage current values measured are less than or equal to those listed in table 4.*

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Table 4.
Admissible permanent values for leakage and auxiliary currents in the patient in mA (milliamperes).

Current path		CF type
	N.C.	
Leakage current to earth	0.5	Leakage current to earth
Leakage current in case	0.1	Leakage current in case
Leakage current in the patient d.c – a.c.	0.01	Leakage current in the patient d.c – a.c.
Leakage current in the patient (mains voltage in applied part)	-----	Leakage current in the patient (mains voltage in applied part)
Auxiliary current in patient d.c – a.c.	0.01	Auxiliary current in patient d.c – a.c.
N.C. = Normal condition S.F.C. = First failure condition		N.C. = Normal condition S.F.C. = First failure condition

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5. Testing the main technical characteristic

Warning: *All tests must be performed in compliance with the provisions of the related general, detailed and performance safety regulations listed in the technical characteristics section*

5.1 Instruments required

a) sample mV generator with the following characteristics:
b) low frequency sine wave generator;
c) ECG simulator.

5.2 Sensitivity test

- ✓ set the device up to record 3 channels on leads V1 to V3 with sensitivity of 20 mm/mV;
- ✓ connect the patient cable to the device;
- ✓ connect terminals C1, C2 and C3 of the patient cable connected to the device to the positive pin of instrument 5.1.a);
- ✓ connect all the other cable terminals to the negative pin of instrument 5.1 ;
- ✓ record the signal for a few seconds;
- ✓ check that the amplitude of the recorded signal is 20mm. +/- 5% on all channels.

5.3 Testing the ECG leads

- ✓ switch the device on;
- ✓ connect the patient cable to the device;
- ✓ connect the red termination of the patient cable to the positive pin of the instrument specified in point 5.a.a and the remaining wires to the negative pin.
- ✓ start recording and check that the amplitude in mm of the signal, and its polarity (positive or negative) comply with the values indicated in table 5.3.
- ✓ repeat the measurements in sequence with the remaining active terminations G - V - C1 - C2 - C3 - C4 - C5 - C6 of the patient cable using

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the method described in c), and check that the values correspond to those indicated in table 5.3.

PATIENT CABLE AND LEADS TEST		
CONNECTIONS FOR THE TEST		
INSTRUMENT		**PATIENT CABLE CONNECTIONS**
Connector	Patient cable terminations	N. 1 Termination to positive apart from black
+ ◎	←	N. 4 Terminations to negative with 5 wire cable
- ◎	↑ → ↓ ° ± ″ ≥ × ∝	N. 9 Terminations to negative with 10 wire cable
Square wave signal from:		Electrocardiograph: amplification 1mV/10mm
1 Hz ± 1%	1 mVpp ± 3%	signal recorded in mm ± 5%

TABLE OF VALUES												
Term. to positi	**LEADS AND VALUE OF IMPULSE**											
	I° mm	**II° mm**	**III° mm**	**aVR mm**	**aVL mm**	**aVF mm**	**V1 mm**	**V2 mm**	**V3 mm**	**V4 mm**	**V5 mm**	**V6 mm**
R	- 10	- 10	0	+ 10	- 5	- 5	- 3,3	- 3,3	- 3,3	- 3,3	- 3,3	- 3,3
G	+ 10	0	- 10	- 5	+ 10	- 5	- 3,3	- 3,3	- 3,3	- 3,3	- 3,3	- 3,3
V	0	+ 10	+ 10	- 5	- 5	+ 10	- 3,3	- 3,3	- 3,3	- 3,3	- 3,3	- 3,3
C1	0	0	0	0	0	0	+ 10	0	0	0	0	0
C2	0	0	0	0	0	0	0	+ 10	0	0	0	0
C3	0	0	0	0	0	0	0	0	+ 10	0	0	0
C4	0	0	0	0	0	0	0	0	0	+ 10	0	0
C5	0	0	0	0	0	0	0	0	0	0	+ 10	0
C6	0	0	0	0	0	0	0	0	0	0	0	+ 10

Table 5.3

5.4 Testing the paper speed

- ✓ switch on the device and connect the patient cable;
- ✓ connect terminals C1 C2 and C3 of the patient cable to the positive pin of instrument 5.1a);
- ✓ connect all the other cable terminals to the negative pin of instrument 5.1a);
- ✓ using the instrument with a square wave of 1 Hz and an amplitude of 1 mVpp;
- ✓ record the signal on leads V1, V2, V3;
- ✓ measure the length of the wave cycle recorded on the paper.

  The results should be as follows:

  Period = 50 mm +/- 5% for a feed rate of 50 mm/s;
  Period = 25 mm +/- 5% for a feed rate of 25 mm/s;
  Period = 5 mm +/-10% for a feed rate of 5 mm/s;

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5.5 Frequency response test

- ✓ switch on the device and connect the patient cable;
- ✓ connect terminals C1, C2, C3 of the patient cable to the positive pin of instrument 5.1.a);
- ✓ connect all the other cable terminals to the negative pin of instrument 5.1a);
- ✓ set the sine wave generator to 10Hz with an amplitude of c. 1mVpp;
- ✓ select leads V1, V2, V3 and a sensitivity of 10 mm/mV.;
- ✓ make a recording and adjust the amplitude of the generator so as to obtain a 10 mm excursion of the signal recorded;
- ✓ vary the generator frequency from 0.5Hz to 100Hz with constant amplitude;
- ✓ check that the frequency response is in accordance with the values in table 5.5.

Amplitude of signal in mVpv	**Unfiltered sinusoidal input signal in Hz**	**Relative tolerance of signal 10 Hz – 10 mm**
1	From 0,67 to 40	± 10%
1	From 40 to 100	+ 10% - 30%
0,5	From 100 to 150	+ 10% - 30%

Table 5.5

Note: *The 0.5 Hz pitch linear phase anti-drift filter is always on and cannot be switched off.*

The 50 or 60 Hz filter eliminates modified notch digital type mains disturbances in the linear phase, with a frequency response of 32 Hz - 3dB.

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6. Functional blocks

6.1 Mother board

The motherboard is based on the following principal components:

Processor

- ✓ 32 bit Fujitsu MB91101 RISC microprocessor with 12 MHz clock
- ✓ The apparatus is supplied moreover of N°1 static memory ram from 4Mb
- ✓ 2 x 8 Mb flash memories including the following software:
  - o *boot code;*
  - o *operating system software;*
  - o *applications software;*
  - o *calibrations data;*
  - o *ECG archive.*
- ✓ Execution of static ram self test.

Connection to mains

Power supply AC/DC type switching, model XPOWER ECM60US18 - 60W

- ✓ Input Voltage : 90-264Vac
- ✓ Input frequency: 47-63Hz

6.2 Battery charger

The battery charger section consists of the following parts:

- ✓ The battery (12V 2100mAh NIMH) is recharge from a circuit to constant current that is under responsibility of Q303 and it develops approximately 1/10 of the nominal current of the battery (150mA).
- ✓ Mains filter against electromagnetic disturbances.
- ✓ Rectification, voltage stabilisation and current limitation circuits.

Testing the battery charger circuit

- ✓ If the mains on led does not light up, check using the following procedure;
- ✓ Disconnect the mains cable;
- ✓ Check the externally accessible mains fuses
- ✓ Check the internal fuse 5A delay SMD
- ✓ Disconnect the battery from the device as indicated in chapter 8.4;
- ✓ Connect an amperometer to the free terminal caps;

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- ✓ Connect the mains cable: the reading should indicate a current of 150 mA ± 15;
- ✓ If the reading is outside the range of values indicated, replace the motherboard.

6.3 Internal circuit power supply

This consists of the following power supplies:

- ✓ +5 V generated by a switching type voltage regulator;
- ✓ VI – VL to supply the patient input analogue circuits on the hybrid circuit (insulated part);
- ✓ + 3.3 V generated by a linear regulator to supply the control logic;
- ✓ + 3 V reference voltage for the A/D converter;
- ✓ VTPH voltage obtained from the battery to supply the thermal head. This voltage is limited to a current of 5A and stabilised in voltage at +25 Vdc enabled only when printing.

Testing internal circuit power supplies

- ✓ Check fuse F1 (T 5A);
- ✓ If no power is supplied to the internal circuits, replace the motherboard.

6.4 ECG acquisition

This consists of the following parts:

- ✓ Patient input connector;
- ✓ Protection against defibrillator discharges;
- ✓ Signal polarisation circuit;
- ✓ Hybrid circuit for amplification, filter and clamp.

6.5 Motor control

This consists of the following parts:

- ✓ Stroboscopic sensor control;
- ✓ Phase comparator between reference frequency and motor feedback;
- ✓ Current amplifier for motor power supply.

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6.6 Display control

This consists of the following parts:

- ✓ Display controller S1D13704 EPSON;
- ✓ Buffer adapter level from 3,3V to 5V;
- ✓ Step up converter for LCD power supply.

6.7 Display assembly

This consists of the following parts:

- ✓ Graphic LCD TFT colour 480x272 pixel, effective area 95 x 54 mm, backlit 9 led (4,3”)

Testing the display

- ✓ Execution of display self-test.

6.8 Battery

NiMH battery 12V-2100 mAh

- ✓ The battery is protected against short circuits by a Polyswitch 3,5A.
- ✓ Complete recharging requires at least 10 hours or longer. The battery can be partially recharged, in this case, to prolong the life of the battery it should be fully discharged and recharged every 2 months.
- ✓ Replace the battery with one of equivalent type, voltage and capacity.

Attention: *In order to ensure a correct equipment operation and safety, replace battery with original Cardioline SpA battery pack.*

Warning: *The battery must only be removed if the device is off and the mains supply cable disconnected. Do not dispose of a spent battery as ordinary refuse or litter.*

Note: *The configuration parameters and any stored ECGs are not lost when the battery is changed.*

Note: *Update the system date and time after replacing the battery.*

Battery check

Proceed as follows to check the efficiency of the battery:

- ✓ Leave the battery on charge for at least 10 hours;
- ✓ Disconnect the mains cable;

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- ✓ Activate printing of an ECG in manual mode on 6 channels at 5 mm/s: if the device shows the battery is flat within the first 5 minutes the battery should be replaced.

6.9 Keyboard

The capacitive keyboard consists of:

- ✓ N° 17 function keys;
- ✓ N° 30 alphanumeric keys;
- ✓ N° 1 LED for mains line indication;
- ✓ CPU connection through bi-directional synchronous serial port.

Testing the keyboard circuits

- ✓ Execution of keyboard self-test;

6.10 Printer unit

This consists of the thermal printer head ROHM model KF2008-GK11B and the mechanical elements required to position it correctly:

- ✓ Effective printing width: 216mm
- ✓ Dot pitch: 0.125mm
- ✓ Total dot number: 1728, 8 dots per mm;
- ✓ support;
- ✓ paper feed unit consisting of 1 direct current reduction gears, equipped with stroboscopic speed control;
- ✓ black mark (page) detection sensor.

Testing the print unit

- ✓ Execution of printer self-test.

6.11 Bluetooth module

The Bt Mitsumi WML-C40 module has the function of transmitting and receiving data from an external PC. Through the BT module it can perform the following functions:

- ✓ Transmission ECG off line (software Cardioline **cube**ecg)*
- ✓ Transmission ECG real time (software Cardioline **cube**ecg)*

* Refer to specific manual for more information.

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6.12 Bridge Usb module

The channel Usb is implemented using the integrated Silicon Labs CP2102, the communication driver usb-serial is downloadable from:
http://www.silabs.com or
http://www.silabs.com/Support%20Documents/Software/CP210x_VCP_Win_XP_S2K3_Vista_7.exe

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7. Troubleshooting

Defect	Possible cause / symptom	Remedy
The device does not switch on in any mode.	• Motherboard / keyboard	Holding down the ON/OFF key, measure 12Vdc on J54 (keyboard connector) pin 3; if absent replace the keyboard card, if not, replace the motherboard.
The device does not switch on when in mains mode.	• Mains LED indicator off.	Check mains fuses, if OK replace motherboard.
The device does not switch on when in battery mode.		• Check T5A fuse on motherboard. • Check efficiency of battery and replace if necessary.
The battery does not charge.	• The device has not been charged for long enough. • Mains LED indicator off. • Defective battery.	• Leave the device charging for at least 18 hours. • Check mains fuses, if OK replace motherboard. • Check efficiency of battery and replace if necessary. • Check T5A fuse on motherboard.
Keyboard keys not working.	• The key pressed not enabled for the specific function.	• Execute keyboard self test and if it fails replace keyboard card.
The paper feeds through without printing.	• Paper not suitable, without black page recognition mark.	• Check that original paper is being used. • Check the paper has been inserted correctly. • Check that the 25 Vdc VTPH power supply to the thermal print head is present when printing is active, if not, replace the mother board. • Check signals on test point table T5 are present. If good, replace thermal head assembly, otherwise replace motherboard.

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Defect	Possible cause / symptom	Remedy
Anomalous printout.		• Clean thermal head. • Execute printer self-test, replace thermal head unit if some dots are missing.
Does not print automatically or does not paginate the trace correctly.	• Paper not suitable, without black page recognition mark.	• Clean mark sensor. • Perform mark sensor calibration. • Replace print unit.
Paper feed defective, or paper finished message with paper present.		• Check paper guides are not damaged. • Clean rubber roller. • Replace print unit.
EC signal disturbed or anomalous.	• Error message on display, on printout.	• See user manual.
The device stops during use.	• No functions operational, no commands accepted.	• Press reset button on right side of device.
Blank display.	• Backlit lamp blown or step up converter failed	• Check for brightness regulation. • Replace display assembly.
No data on display.		• Check for contrast regulation. • Perform display self test. • Verify proper flat connection between mother board and display assembly. • Verify on display connector J48 presence of LCD data signal (level from 0 trough 5 V), if present replace display assembly otherwise replace mother board.

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8. How to disassemble and reassemble the device

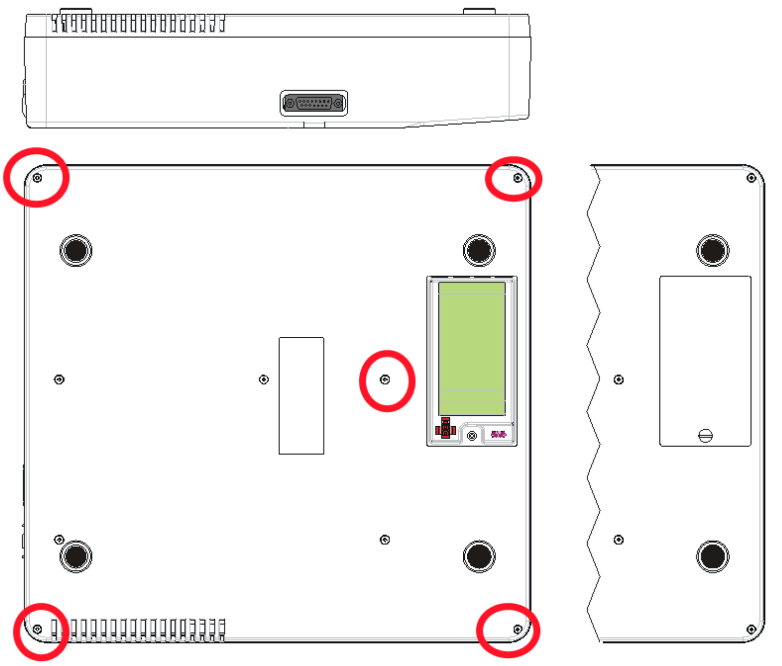
8.1 General precautions

Disconnect the mains cable before opening the unit.
See chapter 15 (procedures for handling ESD components).
To reassemble the device, perform the operations described below in reverse order, ensuring that all subassemblies and connections are performed correctly.

8.2 Opening and closing the device (table T1)

- ✓ Remove the 5 fixing screws from the mobile base;
- ✓ Lift the mobile base of the device;
- ✓ Remove the lower mobile part;

To reassemble the equipment proceed in reverse order.



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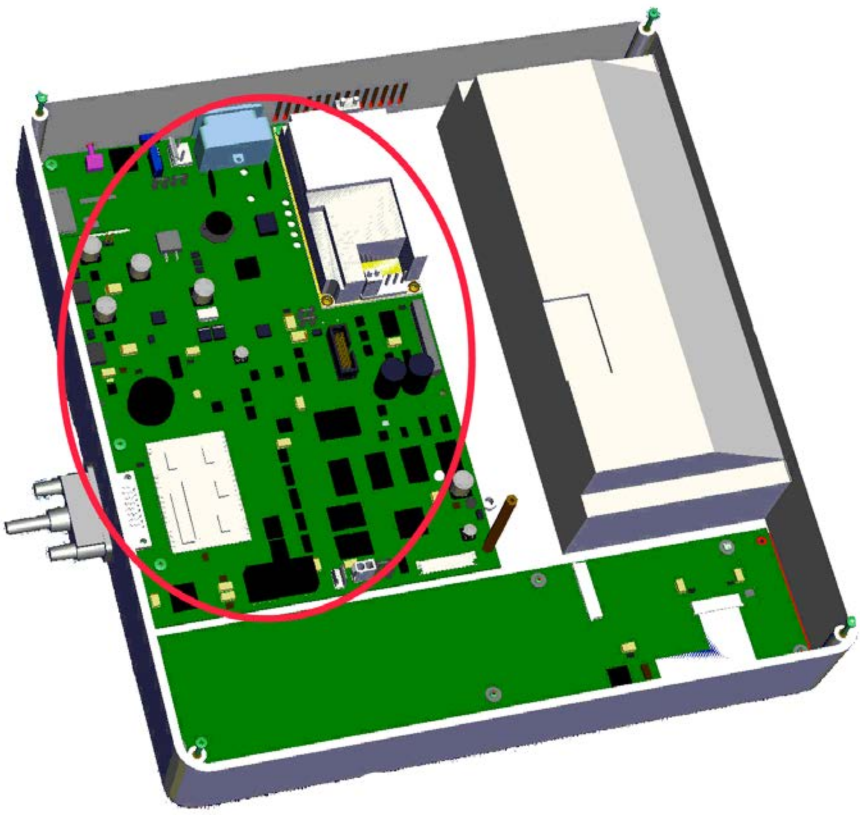
8.3 Removing the mother board (table T2)

- ✓ Proceed following the instructions in chapter 8.2;
- ✓ Remove 2 screws (T2);
- ✓ Lift the patient connector side of the board and disconnect the following:
  - keyboard flat;
  - thermal head flat;
  - motor flat;
  - mark sensor card flat;
  - display connector flat;
  - inverter flat.

To reassemble the motherboard proceed in reverse order.



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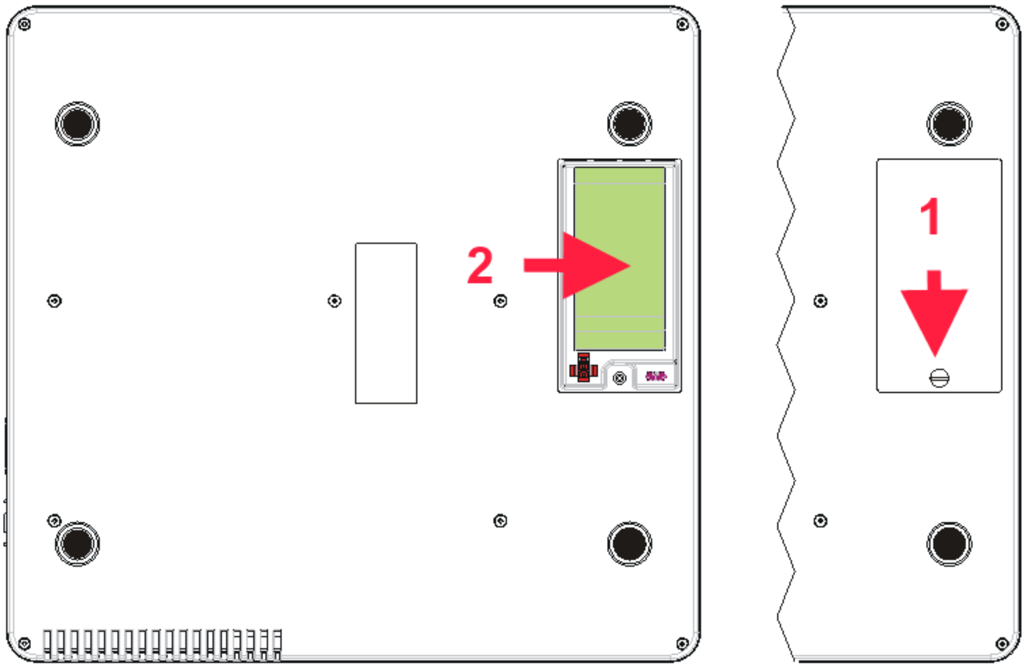
8.4 Removing the battery (table T1)

Proceed as follows:

- ✓ lay the equipment on a soft work surface with the bottom of the casing facing upwards;
- ✓ remove the door of the battery compartment after having slackened its retaining screw (1);
- ✓ disconnect and remove the set of batteries (2);

To reassembly, proceed in inverse order.



Warning: *Respect the correct polarity of the battery connection as specified in Table T1.*

8.5 Removing the keyboard (table T2)

- ✓ Proceed as described in chapter 8.3;
- ✓ Remove 5 screws (T2); replace the keyboard card.

To reassembly, proceed in inverse order.

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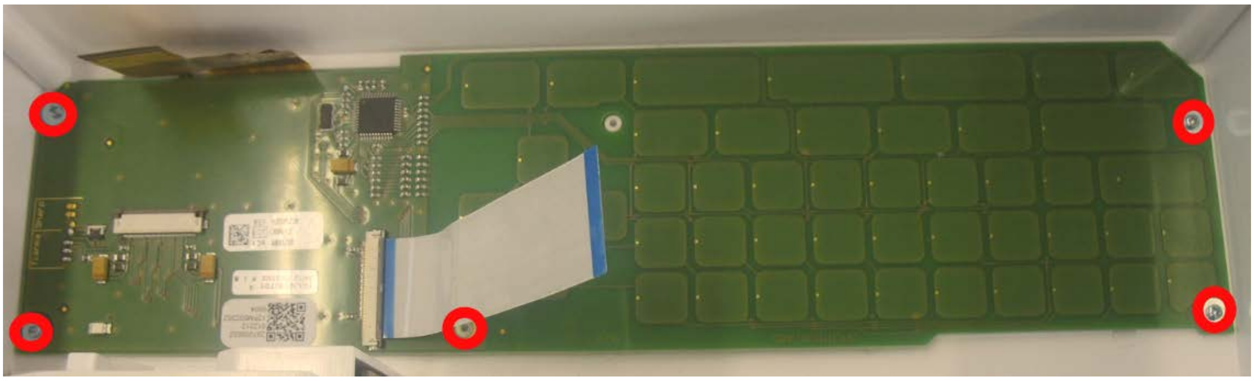

8.6 Removing the display (table T2)

- ✓ Proceed following the instructions in chapter 8.2;
- ✓ Remove the 5 screws attaching the display to the housing (T2);
- ✓ Remove display.

To reassembly, proceed in inverse order.

8.7 Removing the motor assembly (table T2)

- ✓ Proceed as described in chapter 8.3;
- ✓ Remove 2 screws (T2) and remove the motor assembly;

To reassembly, proceed in inverse order.

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8.8 Removing the printer assembly (table T2)

Proceed as follows:

- ✓ proceed as in chapter 8.3 for opening the equipment and removing the mother board from the equipment;
- ✓ remove the 2 retaining screws of the printer assembly, see table T2, remove it from its seat, taking care not to damage the dots of the thermal head with hard objects;
- ✓ proceed in inverse order to fit the printer assembly, paying particular attention to the positioning of the mechanical part with screws, washers, etc.; if the printer assembly is equipped with thermal print head regulation see table T4.
- ✓ to obtain long life of the thermal head, using exclusively the heat-sensitive paper recommended by the manufacturer, packs with height 120 mm is advised. The order code is marked on the bottom edge of the paper;
- ✓ the thermal head is extremely sensitive to electrostatic potentials; it is recommended always to follow the work procedures described in appendix A.

8.9 Removing the mark and paper sensor (table T2)

Proceed as follows:

- ✓ proceed as in chapter 8.3 for opening the equipment and removing the mother board;
- ✓ to remove the sensor board, force it upwards using a suitable tool. It is glued into its compartment inside the upper casing, see table T2;
- ✓ to replace the board it is necessary to clean the area in which it is fitted, position it correctly in its compartment, with the output of the flat towards the paper compartment and stick it down with instant glue;

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- ✓ to close the equipment perform the operations in inverse order;
- ✓ the replacement of the sensor card requires its calibration, see chapter 9.

8.10 Dismantling the paper feed (table T1)

The paper compartment door system is a mechanical device, inserted in the casing, accessible from the outside, driven by the motor of the equipment, which transports the heat-sensitive paper.
It is located above the paper compartment and covers it completely.
To remove this part, proceed as follows:

- ✓ Insert a tool in the special hollow in the left-hand wall of the equipment where the paper comes out and exert pressure upwards so as to release the paper compartment door.
- ✓ This assembly must be disassembled before inserting a roll or pack of paper, when cleaning the roller and always before opening the equipment.
- ✓ To reassembly it, slip it into place with the rubber roller towards the inside of the equipment and holding it up on the other side. As soon as it is inserted, press gently down, on the raised side, so as to snap it shut.

Note: *If the roller is not clean and the paper compartment door is badly inserted or fastened, the paper transport is faulty and the equipment functions incorrectly.*

8.11 Dismantling and replacing the keyboard membrane (table T1)

The keyboard plate is an elastic membrane fitted on top of the keyboard to allow its control buttons to be pressed.
This self-adhesive plate is stuck onto the top of the equipment.
To replace it when worn, lift one corner using a fine blade and pull it off the casing.
If any adhesive is left on the casing, you must remove it by rubbing with your fingertips.
To fit a new plate, centre it with the corners in the space provided and press gently over the whole surface.

Note*: A broken or cracked keyboard membrane compromises the safety of the device.*

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9. Calibration and setting

9.1 General information

This device has a system to automatically set the paper feed speed and mark detection sensor.
The device does not require further setting.
The calibration system may be accessed from the service menu as described below.

Self-test

The device is pre-set to execute self-testing of the main functions: the access sequences for the various menus are guided on the display.
Two types of self-testing may be performed: USER (1) and SERVICE (2), according to the tests to be executed.

User
To enter the self-testing menu, switch on the device, press the MENU key and select and confirm the TOOLS, SELF-TEST, USER(1) submenu using the arrow key (DOWN). The following tests are available in the operator section:

✓ Display: pixel scan. The presence of blank areas signifies faulty operation of the display.

✓ Keyboard: the position of the single keys is simulated in the display. Pressing a given key, the corresponding area of the display is highlighted in reverse. A lack of response in any one area indicates that the relative key is faulty.

✓ Printer: the system prints two triangular waves, the character set in the memory, and signals with different speeds and sensitivities. Irregularities in the printing system are indicated by the presence of non-continuous lines (burnt dots).

✓ Memory: a non-destructive test is performed on the memories (the data in the memory are not cancelled), and a report of the following type is then printed (all tests OK).

<<< ar2100view MEMORY TEST >>>

Application : CRC OK

Ram Memory : OK

✓ Info: the following items of information are printed: model identification, serial number of the device, details of software installed, version and language code, reference of any options installed.

Service
To enter the self-testing menu, switch on the device, press the MENU key and select and confirm the TOOLS, SELF-TEST, SERVICE(2) submenu using the

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arrow key (DOWN). When the access code is requested, press the following keys: FILTER – AMPLITUDES – SPEED – AUTO and confirm.
The following tests are available in the service section:

9.1.1 Paper sensor calibration

To confirm under menu Paper Sensor the Calibration (1), position the mark at a distance from the sensor (visible near the exit from the device), and confirm “next”.

The microprocessor automatically reads the sensor output voltages, voltage with white paper and black mark, and then sets the digital trimmer that regulates the photodiode current to the optimal settings and then stores them.

The paper calibration voltages are shown on the display, as is the numerical position of the digital potentiometer from 0 to 31 positions (wiper pos.).

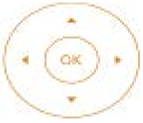
9.1.2 Mark calibration

Confirm the submenu MARK CALIBRATION (1), position the mark at a distance from the sensor (visible near the exit from the device), and confirm “enter”
The microprocessor automatically reads the sensor output voltages, voltage with white paper and black mark, and then sets the digital trimmer that regulates the photodiode current to the optimal settings and then stores them.
The mark calibration voltages are shown on the display, as is the numerical position of the digital potentiometer.

9.1.3 Speed calibration

Testing and setting

Select and firm the SPEED CAL. (2) submenus, and the speed to calibrate
The following messages then appear on the display: Output PWM0 Output – Freq = xxxx Hz with the possibility of selecting from: test – change – exit.
Confirming the “test” command enables the paper feed and the print unit emits a 1 impulse per second signal, if the speed has to be calibrated, select



and increase or decrease the frequency to set the paper feed speed entered. When the speed has been set correctly, select and confirm with “exit”.
The permitted tolerance values for the various paper feed speeds are as follows:

5 mm/s +/- 10%
10 mm/s +/- 10%
12,5 mm/s +/- 10%
25 mm/s +/- 5%
50 mm/s +/- 5%

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9.1.4 Option code

It allows the update of the language and options.

It will involve the loss of all the data which memorized in the instrument. The introduction of a number of 40 figures, released from Cardioline Spa, will qualify the new language and options. Please refers to user manual for instructions on how to active a new option code.

9.1.5 Serial number

Only for factory use.

9.1.6 Print screen

Only for factory use.

9.2 Print head alignment

In case of print head adjustment remove mother board as per chapter 8.3 then follow instructions provided in table T4, repeat adjustment until the best result is obtained.

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10. Periodic maintenance

The **ar**2100**view** electrocardiograph has been designed to assure a high degree of reliability during the life cycle of the product.
Any incorrect conditions of use or anomalous operation will be signalled by messages on paper and on the display.

10.1 Inspection frequency

To guarantee a long and safe duration, the instrument and its accessories must be periodically inspected and checked.
Table 10 indicates the type and frequency of controls required, based on normal use of the electrocardiograph (about 4000 ECG recordings per year).

***Warning**: Check immediately if there are events not referring to normal use.*

General and safety tests

Type of operation	Frequency - months
- full discharging and charging of battery	2
- check and clean printer head	3
- visual inspection of the device, patient cable and accessories	6
- execution of self-test	6
- check and clean paper roller	12
- check paper speed	12
- check keys and keyboard	12
- clean paper compartment and mark presence sensor	12
- check patient cables and electrodes	12
- check whole amplification chain (sensitivity test)	12
- replace battery	12
- check calibrations	12
- check leakage currents	24

Table 10

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11. Cleaning and disinfections

11.1 Cleaning the device, electrodes and patient cable

- *Device:* use a cloth dampened with water or ethyl alcohol. Do not use other chemical products or household detergents.
- *Electrodes:* remove the electrodes from the patient cable and wash under running water. Do not scratch the electrodes and do not wash the leads box and the patient socket.
- *Patient cable:* do not immerse in water, use a cloth dampened with alcohol or an equivalent solvent.

Note: *the device cannot be sterilized! The electrodes can be sterilized with ethylene oxide.*

11.2 Clean thermal head

To clean the printer thermal head correctly use a cloth slightly dampened with alcohol or an equivalent solution with the device off.

Proceed as follows:

✓ open the paper feed;

✓ pass the cloth over the dots of the thermal head without pressing too hard.

Warning: *the printer thermal head is extremely sensitive to electrostatic potentials. Do not touch it for any reason. If necessary, handle it after connecting to a ground via a suitable protective bracelet or belt.*

11.3 Cleaning the paper feed roller

To clean the paper feed roller correctly, remove it and use a soft cloth lightly dampened with alcohol or an equivalent solvent, turning the roller to clean the whole surface.

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11.4 Cleaning the mark and paper sensor

To clean the mark sensor correctly use a cloth slightly dampened with alcohol or an equivalent solution with the device off.

Proceed as follows:

- ✓ remove the paper feed;
- ✓ pass the cloth over mark sensor without pressing.

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12. Spare parts list

The code numbers of the spare parts are listed in table 12.1.
The spare part code is also indicated on the label identifying the main subassemblies inside the device.
To order a spare part, use the corresponding code number.

Spare parts list of Ref 80608064R - AR2100VIEW ECG REC CARDIOLINE

Code	**Description**
69701148	Fuse 5a Smf Slo-blo Smd (10 Pz)
69701360	Mains Socket
69701511	Battery Cover (white)
69701519	Paper Guide ar2100 Cardioline (white)
69701747	Fuse 1,6 At
69701756	Button On-off
69701757	Supply Board Ac/dc Ecm60us18
69701825	Battery Pack 12v 2100mah ar2100view
69701826	Keyboard Membrane ar2100view
69701827	User Interface Board ar2100view
69701828	Motherboard ar2100view
69701829	Usb Board ar2100view
69701830	Display Tft Lcd 480x272 4.3p ar2100view
69701831	Lower Case Complete ar2100view
69701832	Upper Case Complete ar2100view
69701833	Printer Complete ar2100view
69701856	Printer Complete ar2100view IIs, from SN: 0746.13.xx

Table 12.1

***The replacement of the motherboard is supplied complete with firmware in English and basic option.**
To reinstall the configuration before the replacement of the motherboard, the following data must be supplied:

- ✓ device code number (REF)
- ✓ serial number (SN)
- ✓ language
- ✓ options purchased

Cardioline Spa will send a document with the activation code, which will install the correct language and enable the options previously purchased.

***Note:** The data requested can be printed out directly by the instrument, using the info function on the self-test menu.*

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13. Interconnections with medical systems

The device has a Bluetooth - RS232 interface to communicate with a PC equipped with specific programmes, software **cube**ecg.

This type of connection guarantees the insulation required for connection with medical systems.

The communication with a PC occurs through a proprietary protocol.

Cardioline Spa has validated the following Bt interface devices for installation on the usb port of a PC:

- ✓ Conceptronic Bluetooth 2.1 Usb, http://www.conceptronic.net
- ✓ Conceptronic Bluetooth v2.1 USB 2.0 Nano Adapter 200m

Note: do not install the drivers *on the cd rom supplied on the PC. If the drivers should have been inadvertently installed, uninstall them.*

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14. Technical characteristics

A.C. mains power supply	Internal power supply 90-250V 47/63Hz
Maximum current absorbed.....	500 mA a 115 V ~ ±10% 300 mA a 230 V ~ ±10%
Mains protection	Fuse: T 1,6 A 250 V
Internal power source	Rechargeable lead battery NiMH 10x1,2 Vdc; 2100 mAh
Internal power supply protection	Pico fuse SHF SLO-BLO T 5 A Littelfuse
Classification (EN60601-1)......	Device class I
Applied part	CF type
Class (93/42/EEC Directive)	IIa
Defibrillation protection	Internal
Input dynamic.......................	± 300 mV @ 0 Hz.± 10 mV in pass band
Input impedance	>100 MΩ on each electrode
Common mode rejection.........	>94 dB balanced electrode impedance
Frequency response	0,05 ÷ 190 Hz (-3dB)
Time constant	3,3 s
Acquisition	12 bit; 1000 samples/s/channel printing; 500 samples/s/channel in calculation and filters; Resolution (LSB) 5 μV/bit
Leads	12 leads Standard, Cabrera with quality control of the electrodes connection.
Signal memory	10 seconds for each lead in auto isochronous
Recording sensitivity	Manual and automatic: 2,5 - 5 – 10 – 20 mm/mV ± 5%
Writing system	Thermal printer 8 dot/mm Usable print height 210 mm
Print format	3 / 6 / 12 / D3+1 / D3+3 / D6x2 / D12
Paper transport speed	5 – 10 – 12,5 - 25 – 50 mm/s
Thermal paper.......................	Z-Fold pack: page 210x280mm *200FF, gridded
Pacemaker recognition	Recognises pulse in accordance with current IEC standards
Filters..................................	Mains interference: Modified digital notch filter 50–60 Hz linear phase Anti-drift: Digital high-pass 0.5 Hz, linear phase, always enabled Muscular tremors: 3 frequencies: 20, 25 and 35 Hz.
Serial interface	Bluetooth class I
Keyboard	Capacitive with 47 keys

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Display	Visualization 12 ECG channels, functional parameters, HR (30 - 300), included contact electrodes control signal. Colour graphic LCD TFT 480x272 pixels, display effective area 95 x 54 mm, led backlit (4,3”)
Interpretation program	Parameter calculation (optional) ECG interpretation (optional)
Type of use	Continuous
Operating modes	Manual: acquisition and printing in real time Automatic: simultaneous acquisition PC-ECG: real-time acquisition with visualization on PC (optional) Paper Saving: acquisition and automatic archive without printing
Battery capacity	Internal battery: 80 min. printing 6 channels
Recharging time	Internal battery: 10 hours 100%
Housing protection degree	IP 20
Dimensions	325 x 80 x 345 mm (length x height x depth)
Weight	3700 gr. without paper
Operation ambient conditions ..	Ambient temperature: from +10°C to +40°C Relative humidity: from 25% to 95% (without condensation) Atmospheric pressure: from 700hPa to 1060 hPa
Transport and storage ambient conditions.................	Ambient temperature: from -10°C to +40°C Relative humidity: from 10% to 95% (without condensation) Atmospheric pressure: from 500 to 1060 hPa

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15. APPENDIX A

15.1 Procedures for manipulating and storing components (ESD)

All modern electronic components, particularly those based on CMOS technology, may be irreparably damaged by even modest electrostatic discharges.

Precautions must be taken against electrostatic discharges when handling and working on electronic components sensitive to electrostatic discharges. ELECTROSTATIC SENSITIVE DEVICE (ESD).

Staff involved in checking, warehousing, shipping and assembly operations must be earthed through a suitable conductive bracelet compliant with safety standards. If this precaution cannot be taken, the operator must wear suitable antistatic shoes or boots.

Work equipment must be earthed.

Tables, work surfaces and other surfaces on which components are handled must be covered in conductive material and earthed.

All tables and work surfaces must be covered with a layer of conductive material and earthed.

The person performing the repair must be earthed with a suitable protective bracelet compliant with safety regulations. The material must be contained in suitable antistatic bags or containers, and labelled according to MIL STD 129J. The containers must guarantee adequate protection against impact and handling during transport.

All E.S.D. components must be stored in their original boxes and kept in special metal containers. While stored in the warehouse, electronic components must be kept in their original packaging.

Any containers used must be exclusively in metal and/or conductive material. If handled directly, the personnel must adopt the precautions specified above.

During handling cards must be stored in suitable antistatic containers.

Each component sensitive to electrostatic discharges will be identified by the abbreviation ESD.

In the Warehouse area containers are labelled with a suitable symbol.

Follow all the instructions in this procedure when dealing with E.S.D. components.

Warning: *The manufacturer is not responsible for any damage to the device caused by insufficient or inadequate treatment, handling or work methods.*

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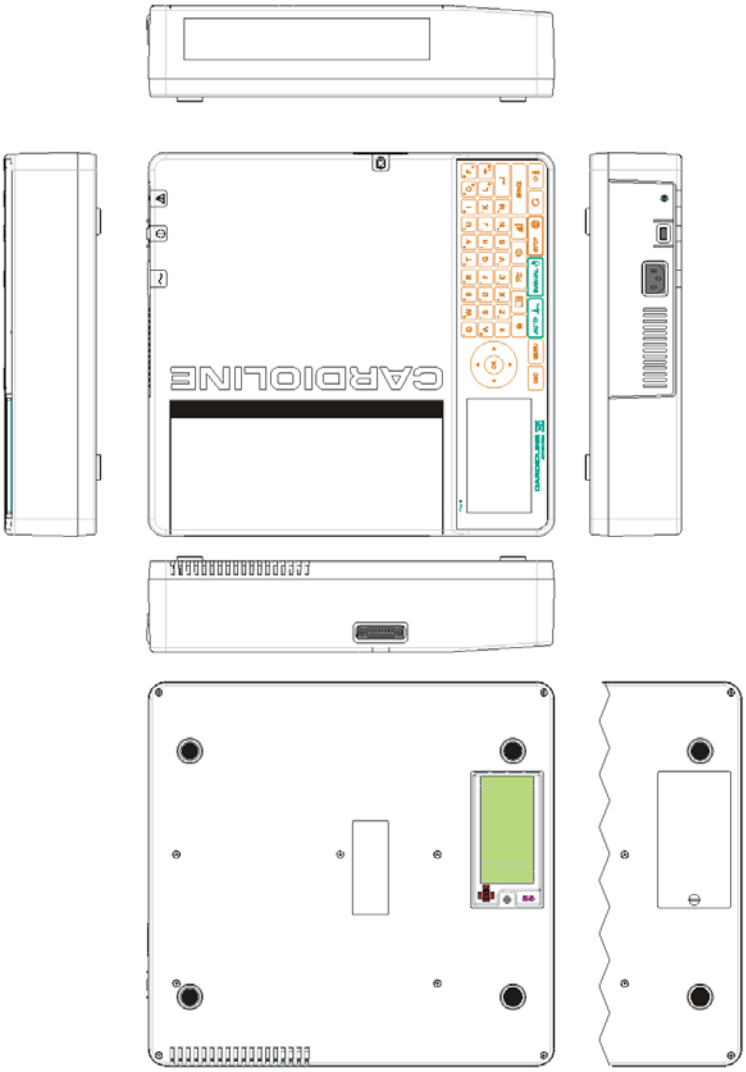
16. APPENDIX B

16.1 Figures and illustrated tables

T1 External view



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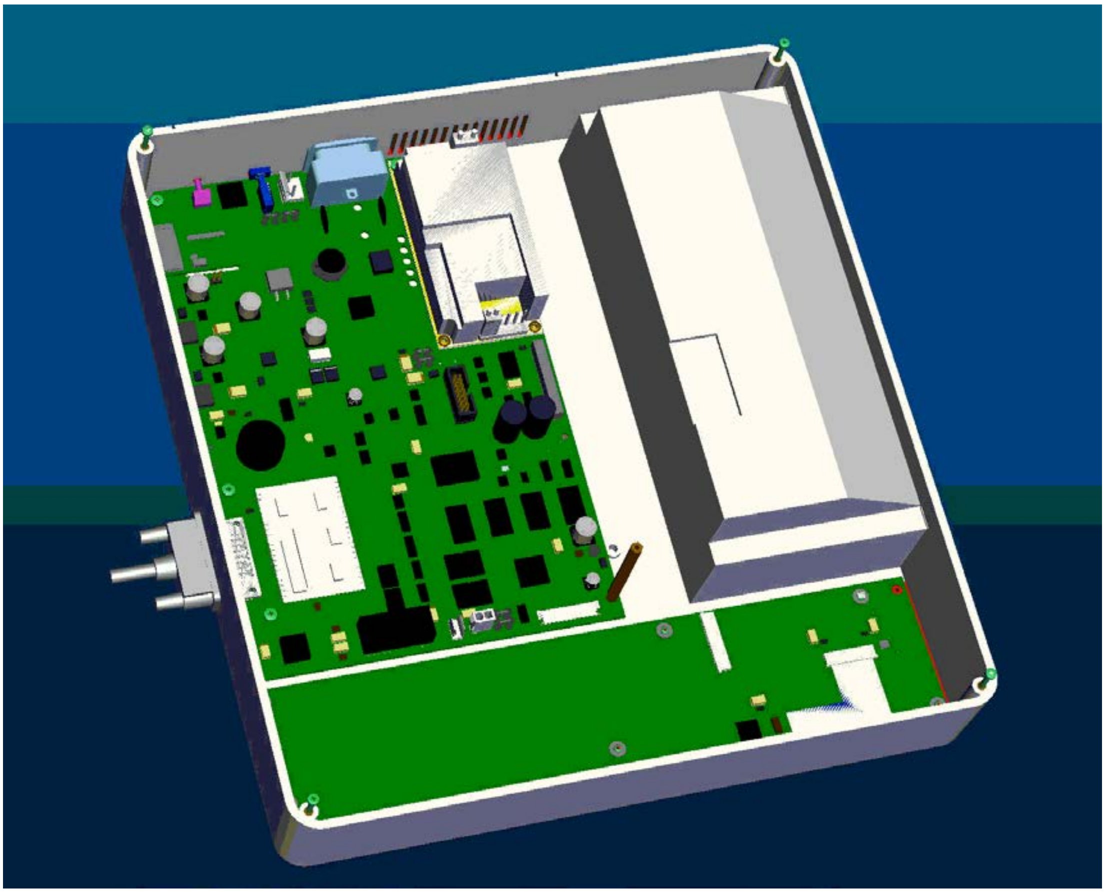
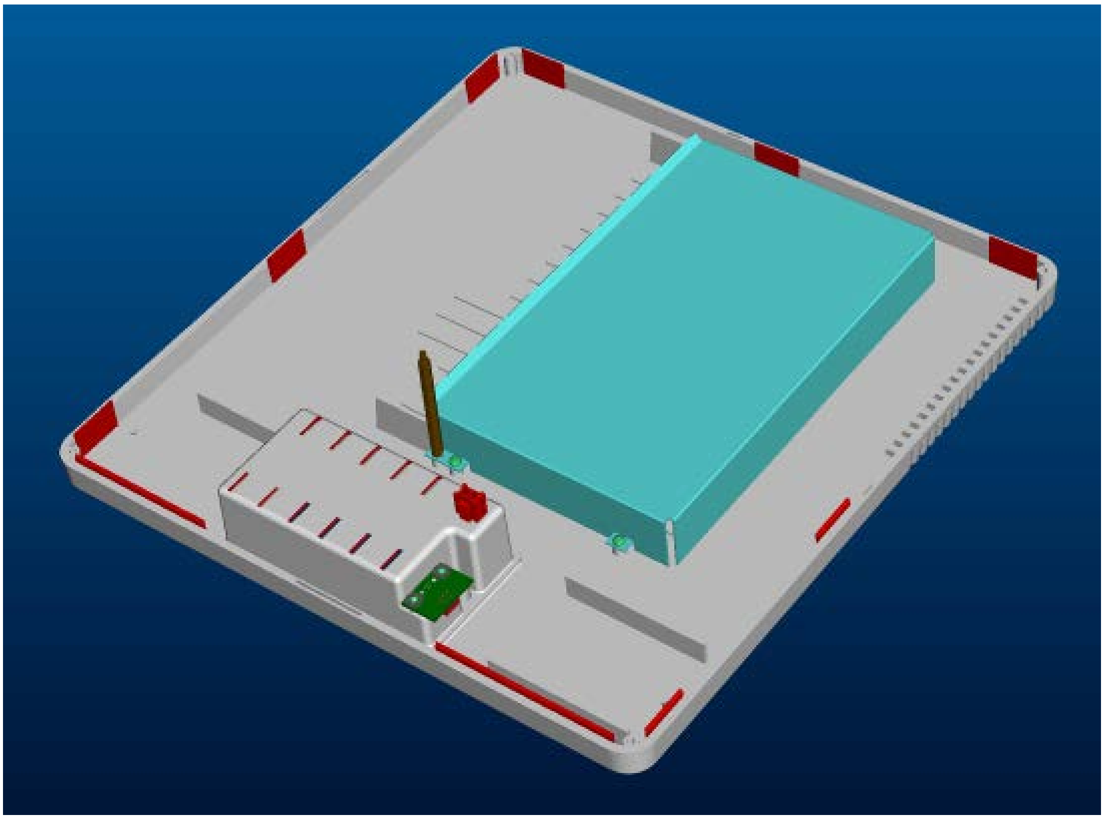
T2 Internal view, fixing and particular of Bluetooth adapter





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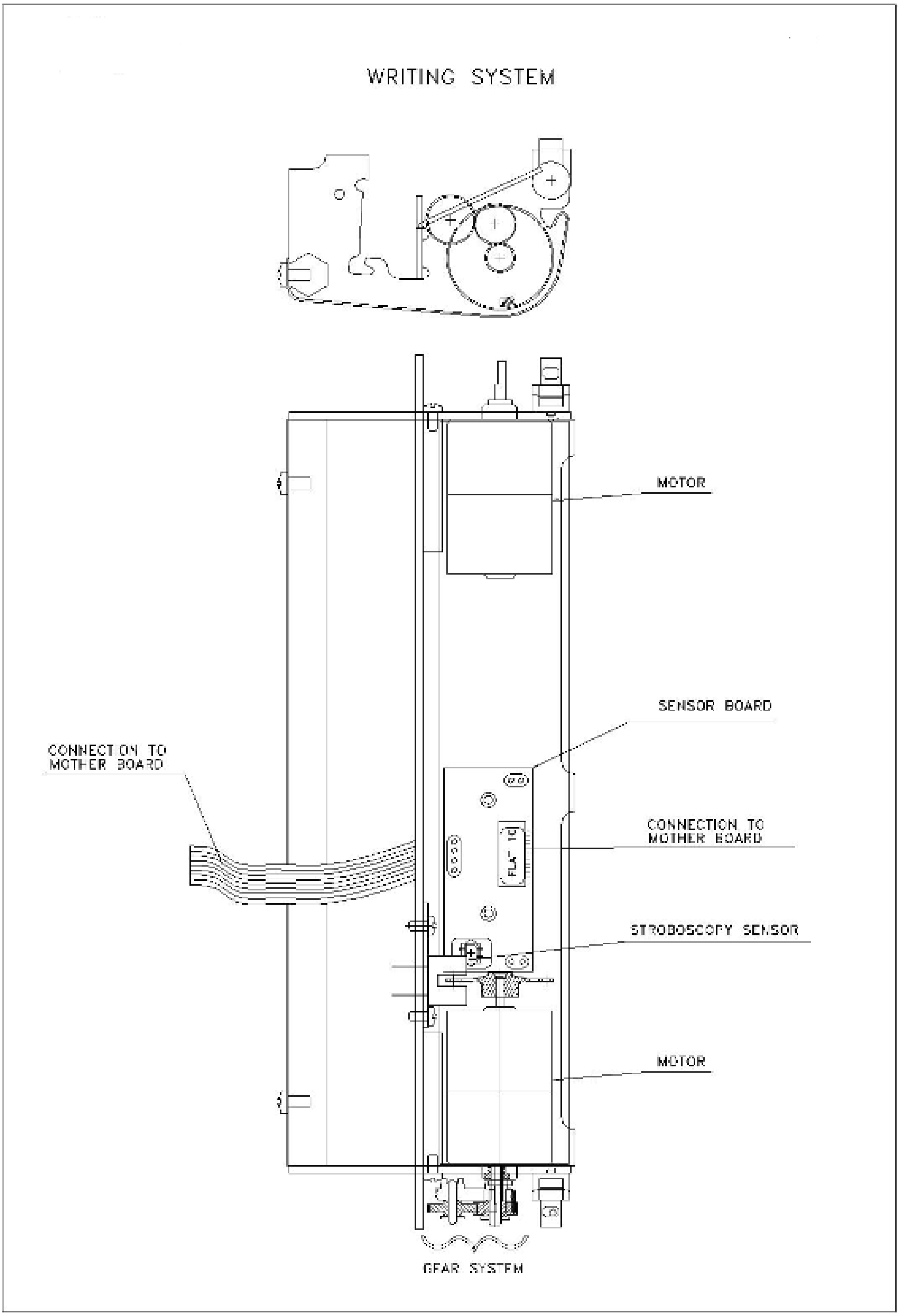
T3 Writing system calibration



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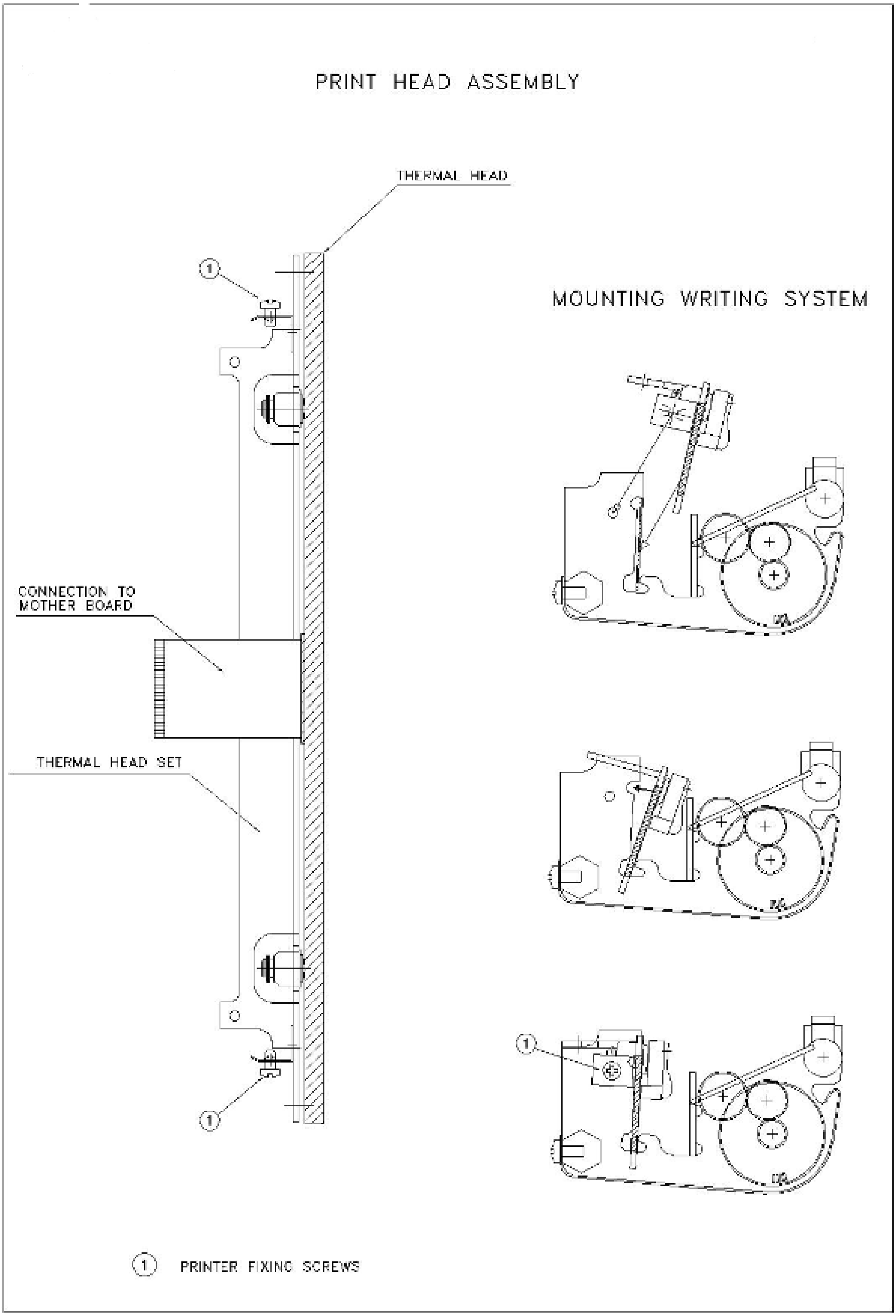

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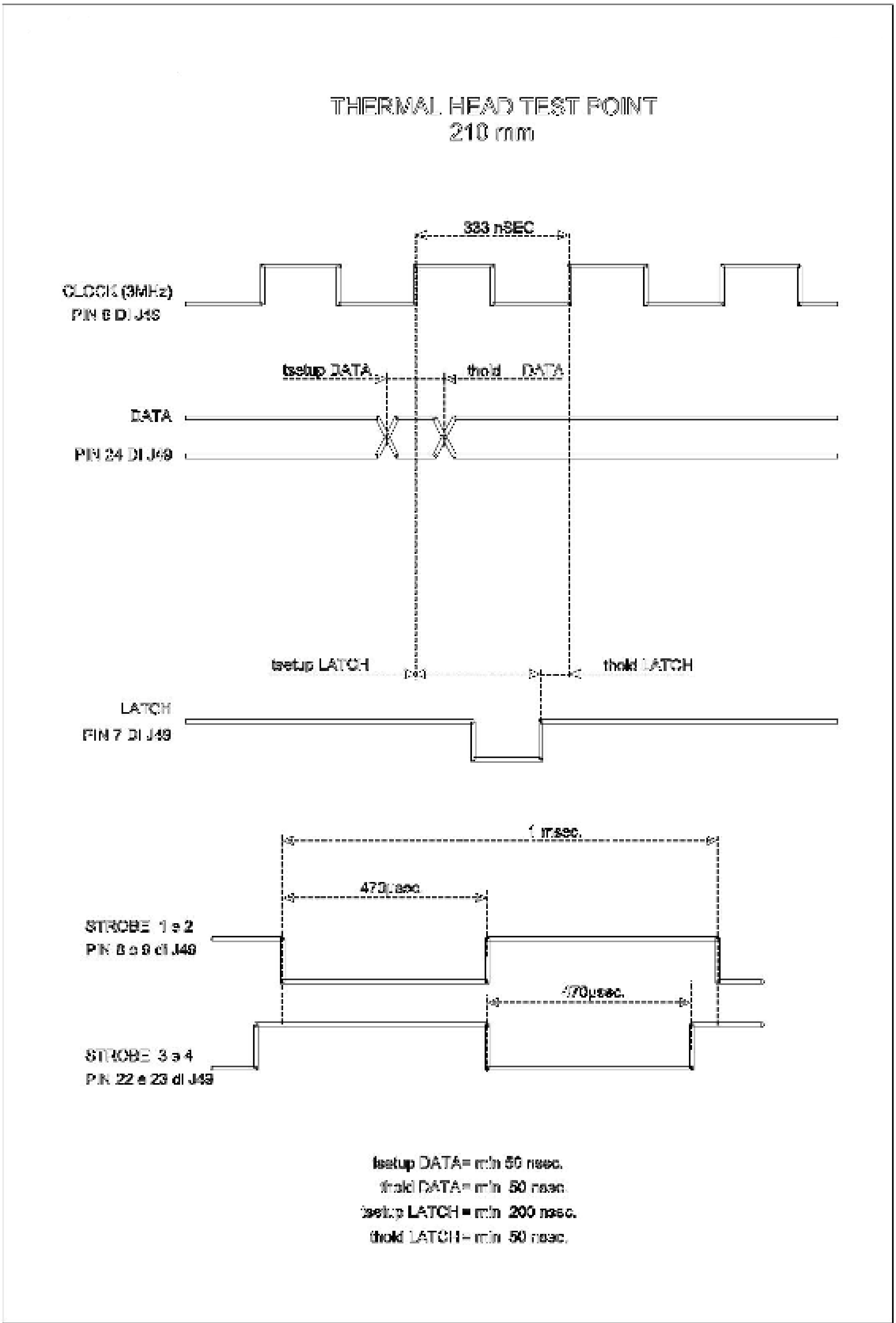
T4 Thermal head test point



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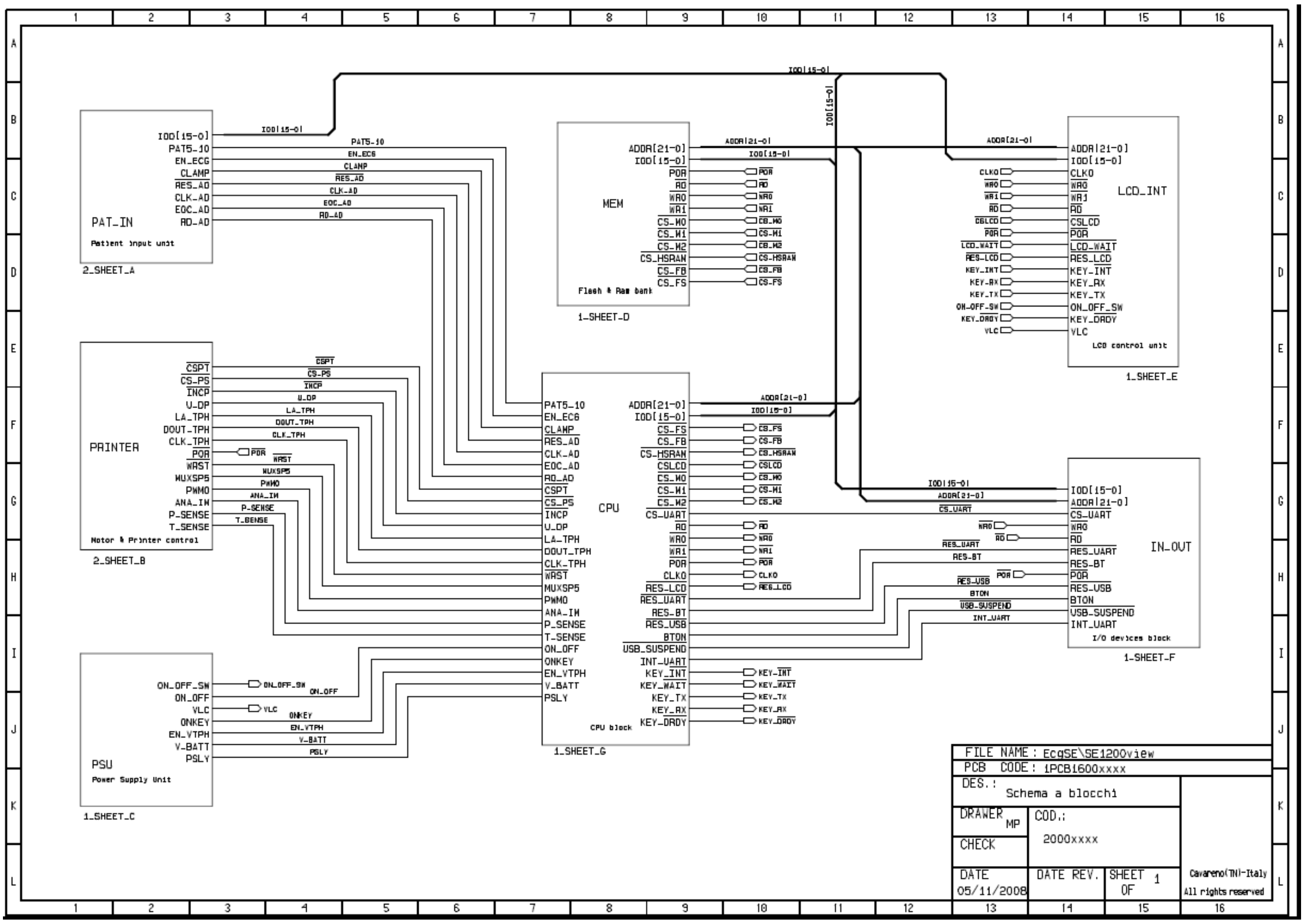
Based on the review of the architecture the following functional blocks are characterized on the Main Board:

1. *PSU*
2. *PAT_IN*
3. *LCD_INT*
4. *IN_OUT*
5. *CPU + MEM*
6. *PRINTER*



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